

General Approach to Solving Spin Moiré Superstructures

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Recently, a host of exciting magnetic textures such as topologically protected skyrmion lattices has been discovered in several bulk metallic lanthanide compounds. In addition to hosting skyrmion phases, a hallmark of this class of materials is the appearance of numerous spin textures characterized by superposition of multiple magnetic modulations: spin moiré superlattices. In order to understand the multitude of complex phases often present in these materials, we require a general-purpose experimental and theoretical framework. Here, we demonstrate such an approach in EuAg_4Sb_2 by comprehensively characterizing and modeling its three complex zero-field magnetic textures. Systematic symmetry-breaking experiments using uniaxial strain determine that the ground-state incommensurate magnetic phase (ICM1) is single q , meaning the magnetic moments modulate along one magnetic propagation vector. In contrast, ICM2 and ICM3 are both double q , meaning they are formed from the superposition of two sinusoidal spin modulations, i.e., spin moiré superlattices. Further, through application of polarized small-angle neutron scattering and spherical neutron polarimetry, we demonstrate that ICM1 is a single- q cycloid and ICM2 and ICM3 are double- q vortex lattices. Despite the quasi-two-dimensional nature of EuAg_4Sb_2 , the modulations propagate out of the ab plane, leading to a shift of the spin texture between triangular lattice planes. Further, the ICM3 to ICM2 transition includes an unusual 45° rotation of the magnetic vortex lattice. Motivated by the coexistence of such drastically different phases in this compound, we conclude by developing a phenomenological model that sheds light on the energetic origins of these varied phases. Our experimental probes and theoretical modeling definitively characterize three different and tunable phases in one material and provide insight for the design of topological spin-texture materials.

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I. INTRODUCTION

Density wave instabilities are ubiquitous in quantum materials and are often intertwined with other emergent phenomena of interest [1,2]. Generally, the presence of such modulations represents a connection between the electronic structure and other degrees of freedom of the

system. In particular, multi- q magnetism formed from the interference of multiple incommensurate magnetic modulations, which are known as spin moiré superlattices (SMSs), are of great interest due to their potential spintronics applications [3]. As these SMSs may, in general, be formed from any combination of potentially low symmetry modulations of varying character, experimental determination of their precise nature can be challenging. Here, using EuAg_4Sb_2 as a model material, we outline a general approach that may be utilized in any material with complex single- q or multi- q incommensurate magnetic (ICM) phases, even in the presence of many coexisting domains.

EuAg_4Sb_2 falls into a family of quasi-two-dimensional (quasi-2D) lanthanide materials that have attracted

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significant recent attention as frustration and conduction electron mediated interactions conspire to drive the formation of a multitude of SMS phases [4–7]. A common observation in this material class is a remarkable number of ICM phases of similar energy, suggesting a nuanced competition of relevant interactions. Additionally, such materials often exhibit a robust coupling between electrical transport and magnetic order—a property especially relevant to potential technological applications in spintronics.

EuAg_4Sb_2 is a rhombohedral, centrosymmetric (space group no. 166, $R\bar{3}m$), quasi-2D material hosting a Eu-triangular lattice with a Eu-Eu distance of 4.7 Å [see Fig. 1(a)]. The Eu layers are separated by 8.2 Å with a Ag_2Sb layer which contributes the conduction electrons. EuAg_4Sb_2 orders at $T_N = 10.7$ K and exhibits three ICM phases upon cooling, which are all also accessible with applied magnetic field $\mathbf{H}||c$ [see Figs. 1(b)–1(e)] [8]. Compared to other recently discovered double- q vortex phase materials such

as $\text{Sr}_3\text{Fe}_2\text{O}_7$ [9], these phases in EuAg_4Sb_2 display a remarkable coupling with the electronic properties, with ICM2 and ICM3 significantly enhancing the magnetoresistance and simultaneously decreasing the Hall conductivity [8,10–12]. This behavior is understood to result from a coherent renormalization of the band structure to a new larger magnetic unit cell [3,8]. The opening of these gaps is critical to understanding the electronic properties of conduction electron mediated magnets and can also be understood as the energetic driver of the stability of these phases [13]. Further, such spin moiré engineering offers a path to the realization of the quantum topological Hall effect [14]. Such behavior motivates the full characterization of each phase, which we pursue herein.

Through a combination of strain control of magnetic domains, polarized neutron measurements, and phenomenological modeling, we reveal that ICM1 is a single- q cycloid [Fig. 1(f)] and ICM2 and ICM3 are two

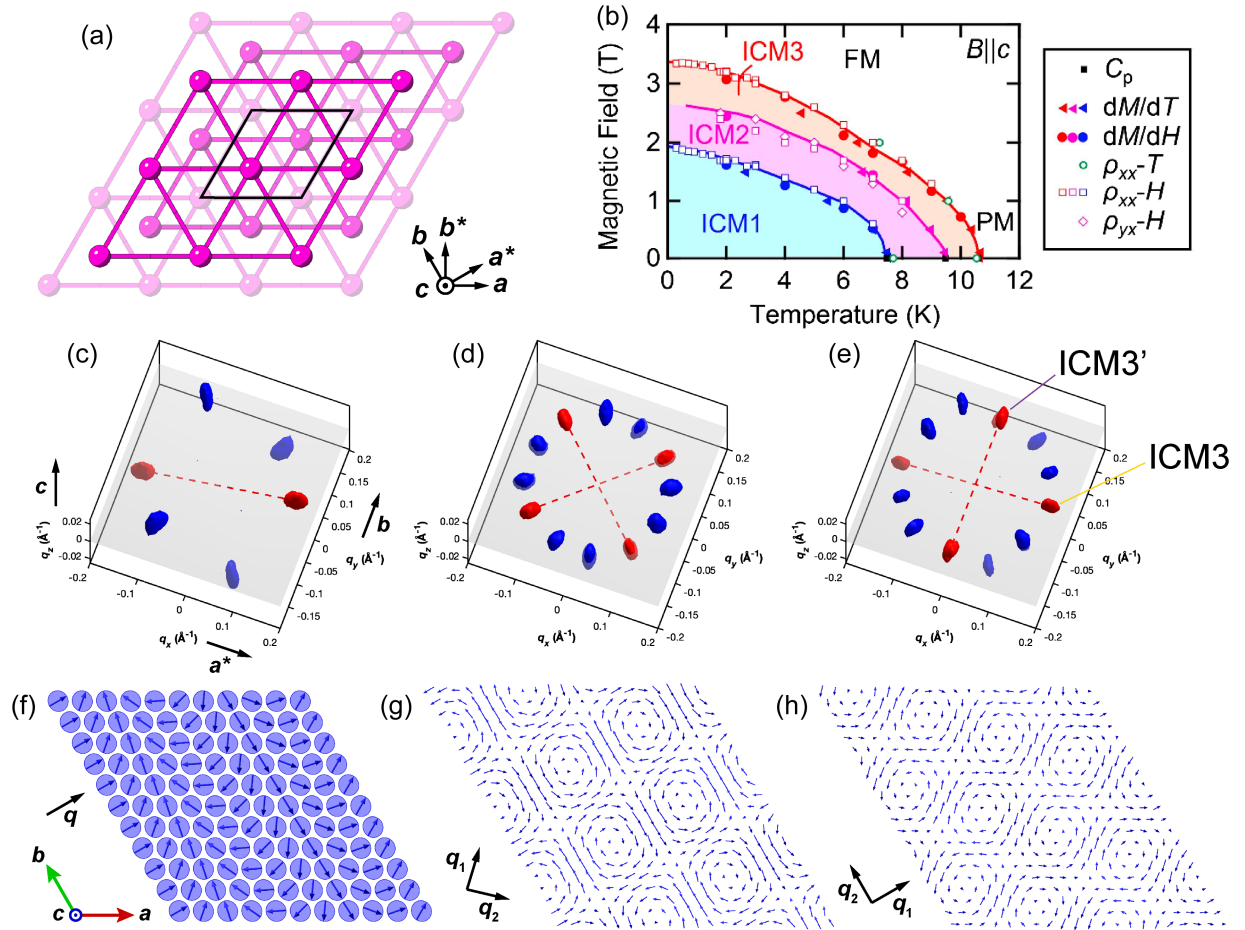


FIG. 1. (a) Rhombohedral ABC stacked europium-triangular lattice layers viewed along the c axis. The unit cell is indicated with a black outline. The reciprocal lattice unit vectors a^* and b^* are indicated in addition to the direct lattice unit vectors. (b) $H||c$ phase diagram for EuAg_4Sb_2 containing the three incommensurate phases ICM1–3, from Ref. [8]. Legend indicates the measurement from which each indicated point was derived. (c)–(e) 3D SANS isosurface plots of ICM1–3 highlighting one magnetic domain each in red. The data are taken in zero field at 2.1, 8, and 9.5 K, respectively. The data are inversion symmetrized and have a high-temperature background subtracted. (f)–(h) Schematic spin structure of ICM1–3. A single Eu layer is shown for simplicity. Propagation direction is also depicted.

inequivalent double- q vortex lattices [see Figs. 1(g) and 1(h)]. ICM1 features a propagation vector (0.105, 0, 0.055), ICM2 features propagation vectors at (0.032, 0.076, 0.010) and (0.108, -0.076, 0.010), and ICM3 features propagation vectors at (0.110, 0.0, 0.037) and (0.06, 0.06, 0.0) in reciprocal lattice units (r.l.u.). Our phenomenological model points to a complex energy landscape that enables the transition between such different phases. The well-defined local moment, anisotropy, and a small four-spin interaction are also found to be important to understanding the phase diagram of EuAg_4Sb_2 .

Our ability to solve the nature and origins of these complex phases with so many low-symmetry domains highlights the power of our multimodal approach in even extreme cases. Such insights advance our understanding of the engineering and control of spin textures in SMS materials, and our demonstration of new tunability such as strain control represents an exciting avenue for the manipulation of multi- q spin textures in technologically relevant materials. Further, the high symmetry of the parent compound and low symmetry of the magnetic propagation vectors lead to a large number of configurational and inversion domains, providing a rich set of addressable states and domain walls that may host distinct functionalities, a potentially advantageous property for spintronics applications [15,16]. These SMS states open a new platform for studying spin textures in metals, with possible applications in spintronic devices through strain or field control [17–19].

II. SOLVING THE MAGNETIC STRUCTURES

Any spin texture in a material with one magnetic site per unit cell (as is the case in EuAg_4Sb_2) can be described as a superposition of spin modulations,

$$\mathbf{M}(\mathbf{r}) = \text{Re} \left[\sum_i \mathbf{M}_i \exp(i\mathbf{q}_i \cdot \mathbf{r}) \right], \quad (1)$$

where i is the index of the magnetic propagation vector $\mathbf{q}_i$, $\mathbf{r}$ is the real-space position, $\mathbf{M}(\mathbf{r})$ is the real-space magnetization density, and $\mathbf{M}_i$ is the complex Fourier amplitude or spin component that characterizes the magnitude, direction, and phase of the i th superimposed spin wave. Uniform magnetization can be included with a $\mathbf{q} = 0$ propagation vector. In order to determine the complete nature of such magnetic textures, it is necessary to answer two key questions. First, which magnetic diffraction peaks belong to which magnetic domains, i.e., which $\mathbf{q}_i$ should be included in Eq. (1). For example, a single- q texture would be striped, while a double- q texture would be a square, rectangular, rhombic, or parallelogram tiling. Here, we will identify the correct set of $\mathbf{q}_i$, the domain structure, by cooling the sample in a strain field. Second, one must determine the spin components ($\mathbf{M}_i$) associated with each propagation vector, that is, what $\mathbf{M}_i$ to use in

Eq. (1). These $\mathbf{M}_i$ define whether the magnetic structure is cycloidal, helical, or another type. These spin components will be determined in the following via polarization analysis. The rich variety of potential spin textures one must distinguish between motivates the need to combine different probes, here *in situ* strain and polarized neutrons, to successfully refine the magnetic structure.

A. Determination of domains and multi- q nature

A concise demonstration of the magnetic domain structure is possible by cooling the sample with applied uniaxial strain, with additional evidence presented in Supplemental Material, Sec. I [20]. The unstrained crystal has a uniform population of domains [see Figs. 2(d)–2(f)]. Upon application of tension (elongating the crystal along the $[-1, 2, 0]$ direction, see Methods), one pair of peaks becomes dominant in ICM1, while two sets of peaks become dominant in ICM2 and ICM3 [see Figs. 2(a)–2(c)]. The application of compression favors the peaks that become weaker under tension [see Figs. 2(g)–2(i)]. The small residual intensity in the minority peaks under tension or compression are likely either due to insufficient or inhomogeneous application of strain or from scattering of the unstrained portion of the crystal held by glue to the jaws. This experiment confirms ICM1 is single q , while ICM2 and ICM3 are double q . ICM2 has nearly perpendicular propagation vectors forming a rhombic lattice, while ICM3 has orthogonal (and symmetry-distinct) propagation vectors forming a rectangular lattice. This observation identifies the appropriate propagation vectors $\mathbf{q}_i$ to use for each phase in Eq. (1). One domain of peaks in ICM1–3 is depicted in red in Figs. 1(c)–1(e), visualized in 3D. Here, we focus on the magnetic domains of one crystallographic twin. Further details of the connection between crystallographic and magnetic domains are found in Supplemental Material, Sec. I Supplemental Material and Ref. [21].

With the modest applied strain (no more than approximately 0.25%), the domains could be controlled by strain cooling, though with more strain it may be possible to switch domains directly within the ordered state. This strain control of magnetic domains suggests that there must be some magnetoelastic coupling that changes the relative energies of the different magnetic domains upon the application of strain. Applying strain along the b axis breaks the threefold rotation symmetry about the c axis. Therefore, the red and white domains depicted in Fig. 2 will still have the same energy, but the black domain may now have a different energy. Which domains are favored will depend in detail on how the magnetic exchange parameters are affected by distortions of the unit cell and interatomic distances.

Strain has been shown to control magnetic domain populations in a variety of compounds [23–26]. In the related EuAl_4 , the skyrmion lattice transitions are accompanied by small (approximately 0.1%) tetragonal-to-orthorhombic structural distortions [27]. If a similar lattice distortion is

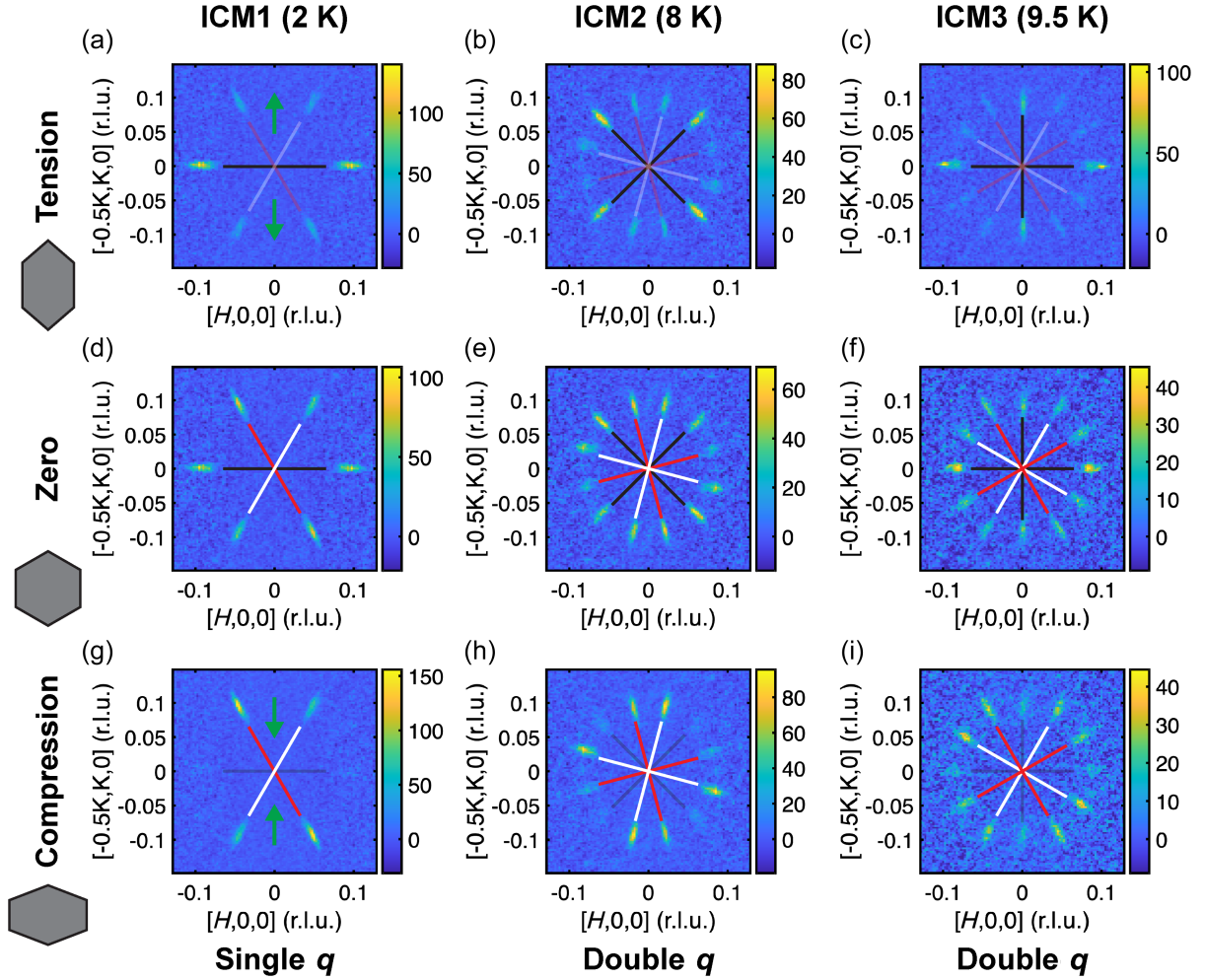


FIG. 2. (a)–(c) SANS pattern measured under tension along the vertical direction [see green arrows in (d)] at 2 K in ICM1, 8 K in ICM2, and 9.5 K in ICM3. (d)–(f) SANS pattern measured with no applied strain (after the crystal broke) under the same conditions. (g)–(i) SANS pattern measured under uniaxial compression along the vertical axis [see green arrows in (f)]. A cartoon (not to scale) of the strained crystal is shown with gray hexagons. The domain structure is indicated with red, white, and black lines. Crossed lines indicate a domain of double- q states. SANS data are inversion symmetrized, smoothed with a half-pixel Gaussian kernel, and have a 15 K background measurement subtracted.

present in EuAg_4Sb_2 as well, as is allowed by the broken threefold rotation symmetry of ICM1–3, then the application of strain will naturally favor the domains that minimize the elastic energy cost of their associated structural distortion. Previous work on Dzyaloshinskii-Moriya interaction- (DMI) stabilized skyrmion compounds MnSi and Cu_2OSeO_3 revealed that uniaxial pressure can align helimagnetic q domains and drastically alter the stability of the skyrmion lattice phase [28–30]. The weaker impact of strain on the magnetic phase boundaries and stability in EuAg_4Sb_2 hints at a difference in the relative stability of the magnetic phases in EuAg_4Sb_2 compared to DMI-stabilized skyrmion compounds.

B. Polarized neutron scattering measurements

To determine the microscopic aspects of the structure, we turn to polarized neutron measurements. The ratio of

spin-flip (SF) and non-spin-flip (NSF) scattering when the polarization axis is aligned with the c axis is depicted in Figs. 3(a) and 3(b). ICM2 and ICM3 are essentially entirely spin flip, indicating that the spin component of the magnetic moment is perpendicular to the polarization axis (the c axis). Therefore, the spins lie in the ab plane within experimental resolution, strongly constraining the allowed textures and indicating that the finite magnetization with applied field originates fully from a uniform ($q = 0$) moment. ICM1 also predominantly lies in the ab plane, but does begin to gain a spin component out of the plane as field is applied along the c axis, from $1.5 \pm 0.2\%$ at $\mu_0 H = 0.05$ T and $T = 2$ K up to $8.7 \pm 0.5\%$ spin-flip scattering at $\mu_0 H = 1.5$ T and $T = 2$ K.

Polarized small-angle neutron scattering (SANS) indicates that the majority of the spin component is within the ab plane and has at least a strong component perpendicular

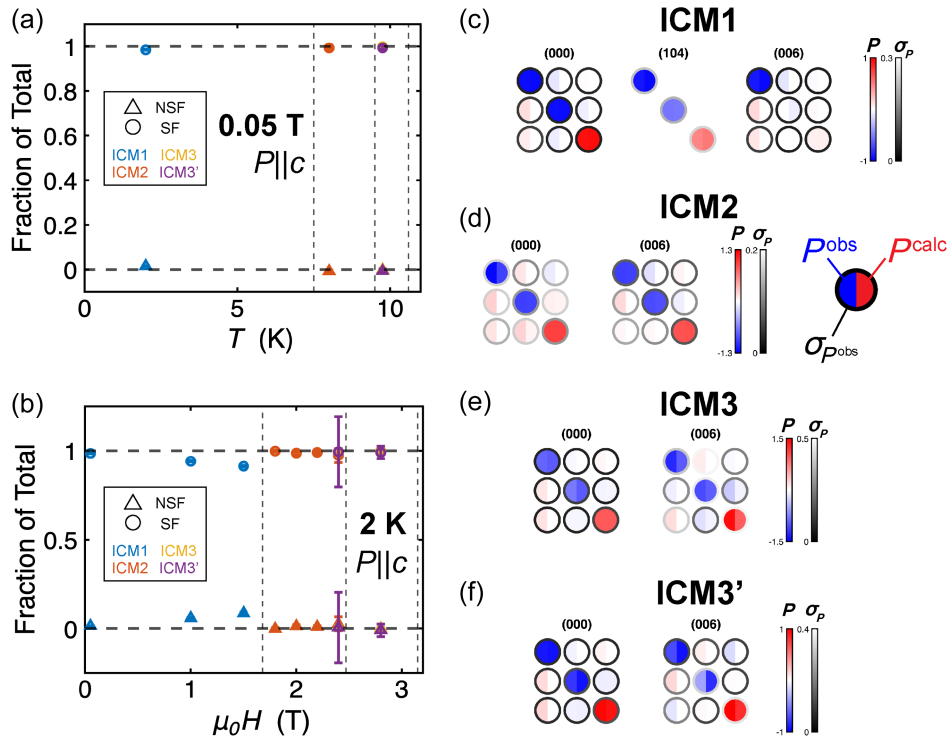


FIG. 3. (a) Polarization-dependent SANS of peaks in 50 mT of field along the c axis as a function of temperature in different phases. (b) Polarization-dependent SANS of peaks at 2 K for various fields along the c axis in different phases. In (a) and (b), the data are background subtracted and corrected for imperfect polarization and analysis efficiency. SF is indicated with circles, while NSF is indicated with triangles. Error bars are the 95% confidence interval from a fit of the peak intensity. ICM3 (yellow) corresponds to the HOL -type peak, while ICM3' (purple) corresponds to the HHL -type peak. (c)–(f) The modeled and observed spherical neutron polarimetry matrices for representative magnetic peaks in ICM1–3. Each matrix is represented with a 3×3 grid of circles. The color (left color bar) of the left side of the circle represents the observed matrix element (P^{obs}), while the color of the right side represents the calculated matrix element (P^{calc}) corresponding to the model presented in Figs. 1(f)–1(h). The color (right color bar) of the outline of each circle indicates the combined statistical and estimated systematic uncertainty (σ_P) of the observed matrix element. The complete set of SNP data is presented in the Appendix.

to the magnetic propagation vector $\mathbf{q}$. However, any spin component along $\mathbf{q}$ (as would be present in a longitudinal spin modulation or cycloid) would not be measurable in SANS, as in SANS the scattering vector $\mathbf{Q}$ is equal to the magnetic propagation vector $\mathbf{q}$. To probe the longitudinal component of ICM1–3, we perform spherical neutron polarimetry (SNP) on the primary propagation vectors in the zeroth zone and on magnetic satellite peaks of the nuclear Bragg peaks. In particular, measurement of the (006) satellite allows a clear probe of the spin component along $\mathbf{q}$ as the scattering vector of the $(006) \pm \mathbf{q}$ peak is nearly perpendicular to $\mathbf{q}$ [see Figs. 3(c)–3(f)].

For each ICM phase, the polarization matrix $P_{\alpha\beta}$ is measured for the symmetry-independent magnetic satellites of the nuclear Bragg peaks. A model of the spin components $\mathbf{M}_i$ of Eq. (1) is fit to the observed polarization matrices for each phase. Observed and calculated matrix elements for several representative magnetic reflections are depicted in Figs. 3(c)–3(f). The full observed and calculated matrices are provided in the Appendix, and further details of the models and fitting are presented in

Supplemental Material, Sec. II [20]. We find that ICM1 is best modeled as a single- q cycloid where the Eu moment rotates in the ab plane [see Fig. 1(f)]. This conclusion is apparent from the fact that the P_{yy} and P_{zz} components diminish toward zero as one nears a scattering vector perpendicular to the propagation vector [Fig. 3(c)], implying spin components in both directions within the plane (see Supplemental Material, Sec. III for additional details on polarization matrix elements [20]). Our model indicates a slight ellipsoidal character where the minor axis along b is 94(1)% as strong as the major axis along a^* , which is consistent with the proposal of Ref. [12].

In ICM2 and ICM3, the diagonal elements remain relatively constant for all observed reflections [see Figs. 3(d)–3(f)], which is consistent with a transverse spin modulation. ICM2 is best modeled as a rhombohedral double- q vortex lattice, where two transverse spin modulations are superimposed to create the vortex lattice in the ab plane [see Fig. 1(g)]. ICM3 is also found to be a double- q rectangular vortex lattice, where two symmetry-distinct transverse spin modulations superimpose to create the

vortex lattice [see Fig. 1(h)]. ICM2 and ICM3 therefore realize a vortex-antivortex crystal with zero net skyrmion number (the unit-cell integral of the scalar spin chirality cancels), i.e., they are topologically trivial in the sense of skyrmion number despite exhibiting strong local chirality when in a finite field.

No significant spin component out of the ab plane was detected to within experimental uncertainty in any of the phases. ICM1 approximately satisfies the equal-moment condition, $|\mathbf{M}(\mathbf{r})| = M_0$, however, the (anti)vortex cores of ICM2 and ICM3 do not. To reconcile this, we propose that, at high temperature, the vortex cores may be either disordered or fluctuating (this higher entropy compared to ICM1 may also explain the stability of these phases at higher temperature in zero field), and at low temperature, as field is applied out-of plane, the cores of the vortices begin to align with the applied field. This effect would produce a small $2q_1$ -type peak intensity in field, which is also observed experimentally (see Supplemental Material, Sec. I and Fig. S3 [20]). The equal-moment condition is strictly valid for fully local, classical spins at zero temperature, but is often a good approximation for Eu^{2+} moments due to the large moment size and strongly localized small orbitals. Indeed, previous work found that the Eu $4f$ orbitals are well below (approximately 1.5 eV) the Fermi energy [8] and show no signs of Eu^{3+} character [31]. Additionally, the Rhodes-Wohlfarth ratio [32] is approximately 1.1, consistent with a local-moment picture. This consideration of the equal-moment condition would explain why ICM1 is favored at lower fields at $T = 0$, while ICM2 and ICM3 are allowed only with applied field at $T = 0$. This picture is also consistent with previous descriptions of this effect in terms of a Landau free-energy picture, where the equal-moment condition leads to a preference for different $\mathbf{M}_i$ to be perpendicular, as in ICM2 and ICM3, or for the components of one $\mathbf{M}_i$ to be perpendicular as in ICM2 [33].

III. PHENOMENOLOGICAL MODEL

Finally, we present a phenomenological model for the magnetism in EuAg_4Sb_2 constructed in momentum space that reproduces the observed ICM ground-state phase diagram as a function of applied field. Such models have been applied to related systems to model phase diagrams with incommensurate order [34]. A generalized momentum-space Hamiltonian is constructed as

$$H = -2J \sum_{\nu, \alpha, \beta} \Gamma_{\nu}^{\alpha\beta} M_{\mathbf{Q}_{\nu}}^{\alpha} M_{-\mathbf{Q}_{\nu}}^{\beta} - H \sum_i M_i^z + H_4, \quad (2)$$

where J is an overall energy scale, Γ is a generalized anisotropic spin interaction in momentum space, $\mathbf{M}$ is the Fourier transformed spin component, α and β are the Cartesian directions, ν is the index of the propagation vector [see included propagation vectors in Fig. 4(a) and

Methods], and i is a site index. Generally, changing which $\Gamma^{\alpha\beta}$ elements are nonzero will shift the preferred ordering between, e.g., a transverse spin modulation, a cycloid, and a helix. The second term in the Hamiltonian is the Zeeman energy term. The third term is a four-spin interaction term H_4 of the form

$$H_4 = -\frac{B}{N} (\mathbf{M}_{\mathbf{Q}_e} \cdot \mathbf{M}_{-\mathbf{Q}_e}) (\mathbf{M}_{\mathbf{Q}_i} \cdot \mathbf{M}_{-\mathbf{Q}_i}), \quad (3)$$

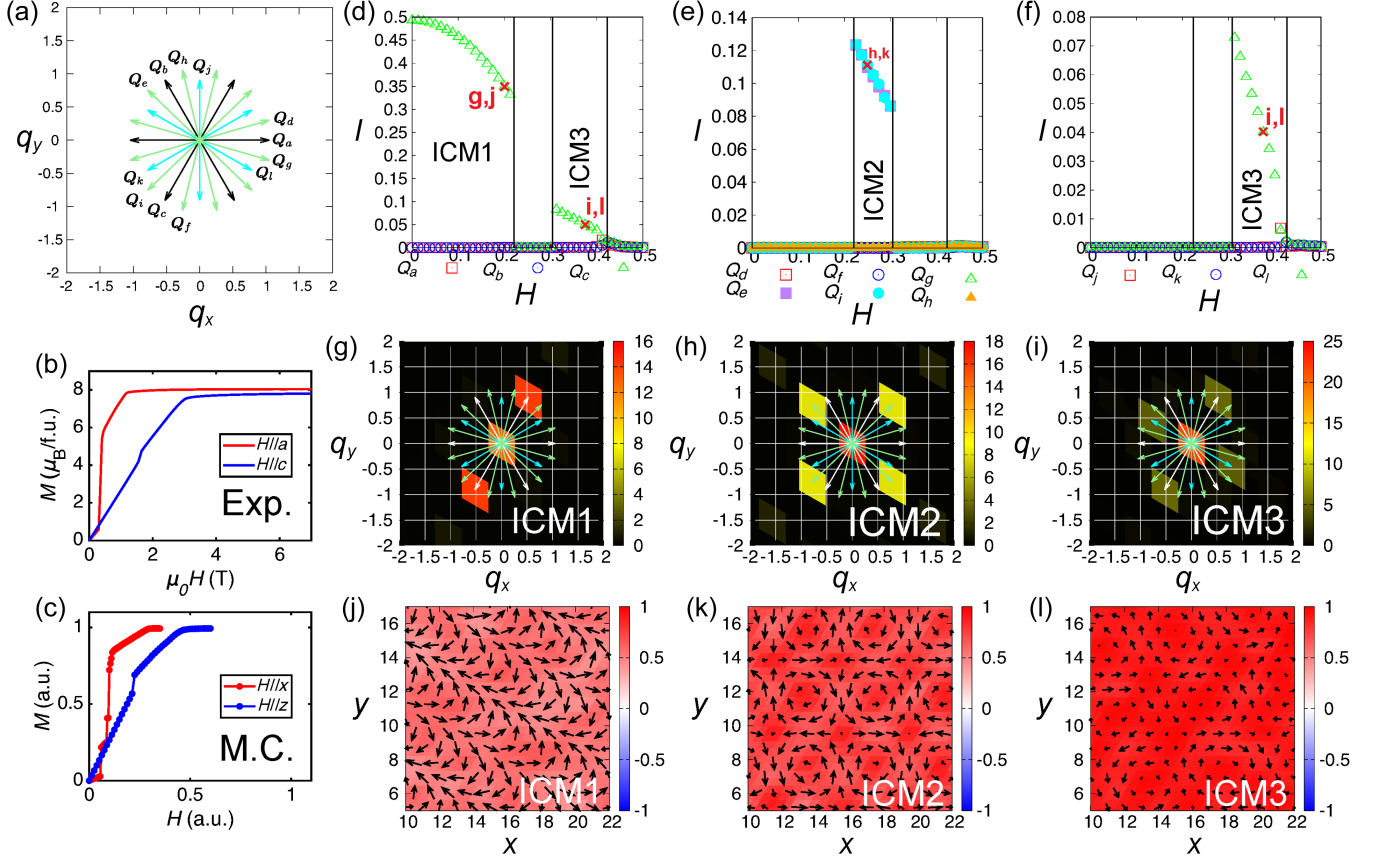
where $\mathbf{Q}_e$ and $\mathbf{Q}_i$ are the two propagation vectors of ICM2 [see Fig. 4(h)], and N represents the total number of spins. We also consider the symmetry-related four-spin interactions.

To stabilize all three phases, we consider three components of Γ : a Heisenberg-type component G along the diagonal, a transverse type that favors a spin component perpendicular to the ordering wave vector, and an anisotropy term D that favors easy-plane anisotropy,

$$\Gamma_{\mathbf{Q}} = \begin{pmatrix} G - A \cos(2\theta) & -A \sin(2\theta) & 0 \\ -A \sin(2\theta) & G + A \cos(2\theta) & 0 \\ 0 & 0 & G - D \end{pmatrix}, \quad (4)$$

where θ is the angle made between $\mathbf{Q}$ and the x axis. The value of $\Gamma_{\mathbf{Q}}$ is, in general, a continuous function of $\mathbf{Q}$, but we consider it only at $\mathbf{Q} = 0$ and the discrete $\mathbf{Q}$ vectors approximately associated with ICM1–3 in this model, and the B -term is included only connecting the ICM2 vectors. Tuning the values of A , G , and B for each of the vectors to $G_0 = 0.9$, $G_1 = 1$, $G_2 = 0.95$, $G_3 = 0.9$, $A_0 = 0$, $A_1 = 0.05$, $A_2 = 0.10$, $A_3 = 0.16$, $B = 0.09$, and $D_i = 0.1G_i$ successfully reproduces the ICM1–3 ground-state phase diagram at $T = 0.01$ as a function of applied out-of-plane field [see Figs. 4(g)–4(i)]. The inclusion of the $\mathbf{Q} = 0$ term causes the model to also qualitatively reproduce the observed anisotropic saturation field [see Figs. 4(b) and 4(c)].

The strength of Γ at all three propagation vectors is similar, which matches with the experimentally observed close energetic proximity of all three phases, being tunable with field or temperature. Additionally, while it is possible to stabilize either ICM1 and ICM3, or ICM1 and ICM2 without it, a four-spin interaction is indeed necessary to explain the simultaneous presence of all three phases, as is the case in other systems [4,35–37]. This four-spin interaction could perhaps originate from the itinerant nature of the present system or the equal-moment condition. One key insight into the stability of the double- q ICM2 and ICM3 states as opposed to a single- q version is that the single- q transverse spin modulations would deviate more from the equal-moment condition than the vortex phases. Simulated annealing at finite temperature stabilizes the ICM3 state, suggesting that ICM3 tends to be more stable at higher temperatures than ICM1 or ICM2. A full simulation of the temperature-dependent phase diagram is complicated by



the first-order nature of the transitions that renders them highly path dependent. Our model also captures the lack of an out-of-plane spin modulation observed experimentally, as well as the elliptical nature of ICM1.

While the isotropic nature of the europium $L = 0$ half filled f -orbital spin state suggests fully isotropic interaction, the finite D_i makes this model slightly anisotropic with an easy-plane anisotropy. Despite the isotropic orbital, in a crystalline environment, the magnetism can become anisotropic. Previous studies of related systems indicate that classical dipole-dipole and spin-orbit coupling with conduction electrons can introduce anisotropy [38,39]. Even isostructural materials can vary significantly in anisotropy due to different spin-orbit coupling and band hybridization [40]. The easy-plane anisotropy of EuAg₄Sb₂ drives a preference for cycloidal and vortex order. Tuning of the anisotropy (perhaps by chemical doping or uniaxial strain) to be easy axis may prefer the formation of

skyrmions. Indeed, the skyrmion hosting EuAl₄ is only weakly anisotropic, which may, in part, explain its tendency toward skyrmion and helical order [41,42].

IV. CONCLUSION

EuAg₄Sb₂ is an exemplary case of how conduction electron-mediated spin interactions can lead to a rich coexistence of spin moiré superlattices that intertwine with electronic degrees of freedom. While previous work in Ref. [8] established the existence of multiple incommensurate phases in EuAg₄Sb₂, here we fully resolve their domain structure and microscopic spin texture via polarized SANS, SNP, and application of *in situ* strain, enabling the first complete phase-by-phase mapping and modeling for this material class. Combined with phenomenological modeling, we are able to reveal both the nature and origins of these complex spin textures—an approach with

immediate relevance across a wide range of materials with exotic spin textures. Further, the demonstration of the coupling of multi- q magnetic domains to strain represents a potentially fruitful route to magnetic domain control. The vortex lattice phases present in EuAg_4Sb_2 therefore offer an ideal platform to study tunable spin moiré superlattices.

In particular, we have demonstrated a domain-selective, polarization-complete neutron strategy for resolving under-constrained incommensurate magnetic textures. By using uniaxial strain to select symmetry-related magnetic domains and comprehensive polarimetry analysis to determine vector chirality, we (i) distinguish single- q cycloidal order from double- q vortex-antivortex spin moiré superlattices, (ii) resolve their phase connectivity, including the lattice reorientation between adjacent vortex states, and (iii) show that the minimal phenomenological model requires higher-order interactions to stabilize the observed multi- q sequence and its anisotropic saturation behavior. Beyond EuAg_4Sb_2 , this combined domain-selective, polarimetry approach is broadly applicable to multi- q magnets where standard neutron diffraction measurements of peak positions and intensities alone are clearly insufficient to uniquely determine the real-space textures.

Further, the demonstration of a coupling between magnetic domains and symmetry-breaking uniaxial strain represents a potentially fruitful route to the control of multi- q magnetic domains. The vortex lattice phases present in EuAg_4Sb_2 therefore offer an ideal platform to study strain-tunable spin moiré superlattices.

We also note that the system displays an easy-plane anisotropy, where all three phases prefer modulated textures with spin components predominantly within the ab plane. This anisotropy is consistent with the lower in-plane saturation field observed in magnetization measurements. If the easy-plane anisotropy is lifted, the system may instead favor a skyrmion or meron-antimeron lattice state. This insight suggests that probing bulk magnetic anisotropy can provide insight into what phases may be present in a frustrated system before the application of more sophisticated techniques to determine the precise phases. This observation also motivates targeted experiments such as in-plane field experiments [43] and chemical or physical strain tuning of interlayer thickness that may tune the system into a phase with nonzero skyrmion number. Exploring the tunability of the phase diagram would further test our model parameters, providing a predictive loop between experiment and theory.

Consideration of the equal-moment condition also justifies why the ground state at zero field is a single- q cycloid, while SMS of vortex lattice type become favorable at elevated temperatures or with applied field at $T = 0$. The ground-state ICM1 completely conserves the saturated moment, and ICM2 and ICM3 only have the spin go to zero at pointlike (anti)vortex core centers, whereas a transverse spin modulation contains lines where the spin

goes to zero. At higher temperatures in ICM2 and ICM3, the vortex and antivortex cores may be thermally fluctuating (stabilizing the phases via their greater entropy than ICM1), while at low temperature and high field they may become field polarized to satisfy the equal-moment condition, explaining the small $2q_1$ -type peaks observed (see Supplemental Material, Sec. I [20]). These observations provide insight into the design principles of multi- q materials: The magnetic moment of Eu^{2+} is relatively isotropic (being an $L = 0$ state) and local in character, so the moment is largely Heisenberg-like and favors any spin modulation that respects the equal-moment condition. The addition of frustration and conduction electron-mediated interactions further leads to a plethora of tunable spin moiré superlattice states. These insights extend to the engineering of SMS phases in any lanthanide metal that exhibits signs of frustration.

V. METHODS

A. Sample synthesis

Single crystals of EuAg_4Sb_2 were synthesized via a self-flux method described previously in Ref. [8]. The crystals exhibit clear hexagonal or triangular platelet morphology enabling straightforward alignment, with characteristic crystal sizes of order $4 \times 4 \times 0.1 \text{ mm}^3$.

B. Small-angle neutron scattering

SANS was performed at the Institut Laue-Langevin on the D33 beamline, at the High Flux Isotope Reactor on the GP-SANS beamline at Oak Ridge National Laboratory, and the Swiss Spallation Neutron Source on the SANS-I beamline at the Paul Scherrer Institut. Neutrons of wavelength 3.1–5 Å were used. The sample was rocked about the vertical and/or horizontal axes to map a 3D volume of reciprocal space. SANS measurements were analyzed in GRASP [44] and in the GRIP module [45]. Every measurement has a high-temperature background subtracted.

C. Strain SANS

For the uniaxial strain SANS experiment, measurements were performed on SANS-I at PSI using a Razorbill UC220T piezoelectrically driven strain cell with clearance for a transmission scattering measurement on a bespoke sample stick. The strain cell is equipped with calibrated capacitive sensors that measure the displacement and force applied to the crystal. This cell can apply up to 50 N or 25 μm of strain (depending on the Young's modulus and length of the sample), with a standard sample size of $\sim 0.1 \times 1 \times 3 \text{ mm}^3$. To reduce beam absorption by the europium, 3.1 Å neutrons were used. A small portion of sample in the titanium strain cell jaws was relatively unstrained; small spot-size measurements confirmed that less domain imbalance was present in this region.

Unpolarized SANS measurements were analyzed in GRASP [44] and in the GRIP module [45].

D. Polarized SANS

Polarization analysis SANS [44,46,47] was performed on the D33 beamline at the Institut Laue-Langevin in Grenoble, France. The incident beam is spin polarized along the vertical axis by an FeSi supermirror polarizer and can be flipped with an rf spin flipper. The outgoing neutron spin is measured with a cell of gaseous ^3He , where the nuclear spins are polarized to approximately 78%. The neutrons are selectively absorbed by the ^3He when their spin is antialigned with the ^3He nuclear spin [48,49]. The polarization of the ^3He is maintained with a constant field chamber and can be flipped through the application of an rf pulse in a so-called “magic box.” In this experiment, one quadrant of the SANS pattern was fully covered by the ^3He , allowing access to all symmetry-distinct peaks. Additionally, the polarization efficiency was monitored by regularly measuring the intensity of the direct beam in both the spin-flip and non-spin-flip channels throughout the course of the experiment to characterize the depolarization of the ^3He cell and allow for correction of the imperfect polarization. The scattering intensity was then corrected for nonmagnetic background and an imperfect flipping ratio to the true polarization-dependent scattering cross sections $d\sigma^{\pm\pm}/d\Omega$ and $d\sigma^{\pm\mp}/d\Omega$ [44,50].

We used 4.7 Å neutrons, and the ^3He cell was filled to 0.8 bar with a 10-cm cell length, resulting in an estimated transmission of 28% and an estimated filtering efficiency of 97%. This setup resulted in a total polarization analysis efficiency starting at $\sim 90\%$, which decayed with a time constant of ~ 200 h. In this geometry, the polarization-dependent magnetic scattering (neglecting all nuclear and nuclear-magnetic terms) is given by

$$\frac{d\sigma^{\pm\pm}}{d\Omega} \propto |M_{\perp,z}^2| \quad (5)$$

and

$$\frac{d\sigma^{\pm\mp}}{d\Omega} \propto |M_{\perp,x}^2| + |M_{\perp,y}^2| \mp i(M_{\perp,x}M_{\perp,y}^* - M_{\perp,y}M_{\perp,x}^*). \quad (6)$$

$M_{\perp}(\mathbf{Q}) = \hat{\mathbf{Q}} \times \{\mathbf{M}(\mathbf{Q}) \times \hat{\mathbf{Q}}\}$ is the component of $\mathbf{M}(\mathbf{Q})$ perpendicular to $\mathbf{Q}$, and x , y , and z are the conventional polarization coordinates [51]. Hence, the SF channel scattering quantifies the component of modulated magnetization in the plane perpendicular to $\mathbf{Q}$ (a transverse in-plane spin modulation), and the NSF channel quantifies the component of modulated magnetization out of the plane along c . Additionally, the difference between $d\sigma^{+-}/d\Omega$ and $d\sigma^{-+}/d\Omega$ quantifies the chiral term that measures to what degree $M_{\perp,x}$ and $M_{\perp,y}$ are out of phase with one another.

Additional details of the polarization analysis may be found in Supplemental Material [20].

E. Spherical neutron polarimetry

Because of the relatively small wave vector of the ICM phases, a cold neutron instrument is ideal to observe the magnetic peaks in EuAg_4Sb_2 . Therefore, we employed the cold triple-axis spectrometer ThALES at the Institut Laue-Langevin in Grenoble, France, which can be equipped with the cryogenic polarization analysis device (Cryopad) spherical neutron polarimeter [52]. Analysis and visualization of SNP data were performed with Mag2Pol [53]. Both statistical and estimated systematic uncertainties (following Ref. [54]) were included. The sample was manually reloaded multiple times with different ψ (the angle between the scattering plane and the a^* axis) to place the different magnetic peaks of interest in the horizontal scattering plane of the instrument: $\psi = 0^\circ$ for the (*HOL*) peaks of ICM1 and ICM3, $\psi \approx 13.5^\circ$ for the ICM2 peaks, and $\psi = 30^\circ$ for the (*HHL*) ICM3 peaks. A beam collimator and 5 Å neutrons were employed to provide sufficient resolution to observe the magnetic peaks of interest. The principle peaks in the zeroth zone appear at a 2θ of 7.3° from the direct beam, which is a large enough angle to measure reliably despite the elevated background from the direct beam. In this configuration, the magnetic satellites can be observed about the (000), (003), and (006) peaks for all ψ , and the (101), (102), (104), and (105) peaks can be accessed for the $\psi = 0$ measurement in the (*HOL*) scattering plane. The satellite peaks observed about the (003) and (006) sum together the propagation vector from both crystallographic twins due to the broader instrument resolution along $\mathbf{Q}$. Additional details of the polarization analysis may be found in Supplemental Material [20] and Ref. [22].

F. Phenomenological model

For the momentum-resolved interaction in Eq. (2), we adopt the following wave vectors: $\mathbf{Q}_0 = (0, 0)$, $\mathbf{Q}_a = Q(1, 0)$, $\mathbf{Q}_b = Q(-1/2, \sqrt{3}/2)$, $\mathbf{Q}_c = Q(-1/2, -\sqrt{3}/2)$, $\mathbf{Q}_d = (Q, Q')$, $\mathbf{Q}_e = (-Q/2 - \sqrt{3}Q'/2, \sqrt{3}Q/2 - Q'/2)$, $\mathbf{Q}_f = (-Q/2 + \sqrt{3}Q'/2, -\sqrt{3}Q/2 - Q'/2)$, $\mathbf{Q}_g = (Q, -Q')$, $\mathbf{Q}_h = (-Q/2 + \sqrt{3}Q'/2, \sqrt{3}Q/2 + Q'/2)$, $\mathbf{Q}_i = (-Q/2 - \sqrt{3}Q'/2, -\sqrt{3}Q/2 + Q'/2)$, $\mathbf{Q}_j = Q''(0, 1)$, $\mathbf{Q}_k = Q''(-\sqrt{3}/2, -1/2)$, and $\mathbf{Q}_l = Q''(\sqrt{3}/2, -1/2)$ with $Q = \pi/3$, $Q' = \sqrt{3}\pi/18$, and $Q'' = \sqrt{3}\pi/6$, where we consider the interaction parameters $G_0 = 0.9$ and $A_0 = 0$ for $\mathbf{Q}_0$, $G_1 = 1$ and $A_1 = 0.05$ for $(\mathbf{Q}_a, \mathbf{Q}_b, \mathbf{Q}_c)$, $G_2 = 0.95$ and $A_2 = 0.1$ for $(\mathbf{Q}_d, \mathbf{Q}_e, \mathbf{Q}_f, \mathbf{Q}_g, \mathbf{Q}_h, \mathbf{Q}_i)$, and $G_3 = 0.9$ and $A_3 = 0.16$ for $(\mathbf{Q}_j, \mathbf{Q}_k, \mathbf{Q}_l)$. The anisotropic term D_i is set to $0.1 \times G_i$ for all wave vectors.

The optimized spin configurations for the model are obtained through an iterative simulated annealing scheme

combined with single-spin updates based on the Metropolis Monte Carlo algorithm. Simulations assume the equal-moment condition, which is valid at the low temperature $T = 0.01$ simulated. Periodic boundary conditions were imposed throughout, and simulations were performed for multiple lattice sizes to verify the robustness and convergence of the numerical procedure.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [56].

APPENDIX: SNP DATA

The polarization matrix elements used in our models are presented here for all three phases. All polarization elements for all phases are visualized in Fig. 5. The numerical values for ICM1 are listed in Table I, ICM2 in Table II, and ICM3 in Table III for the $H0L$ peak, and Table IV for the HHL peak. All the data presented here have a high-temperature background subtracted to isolate the magnetic signal, but are not corrected for the flipping ratio, which was approximately 40. All error bars include both statistical and estimated systematic uncertainties. Discrepancies between the model and the data may be due to a combination of statistical fluctuations, systematic errors due to the imperfect manipulation of the polarization vector in Cryopad, or unequal domain population.

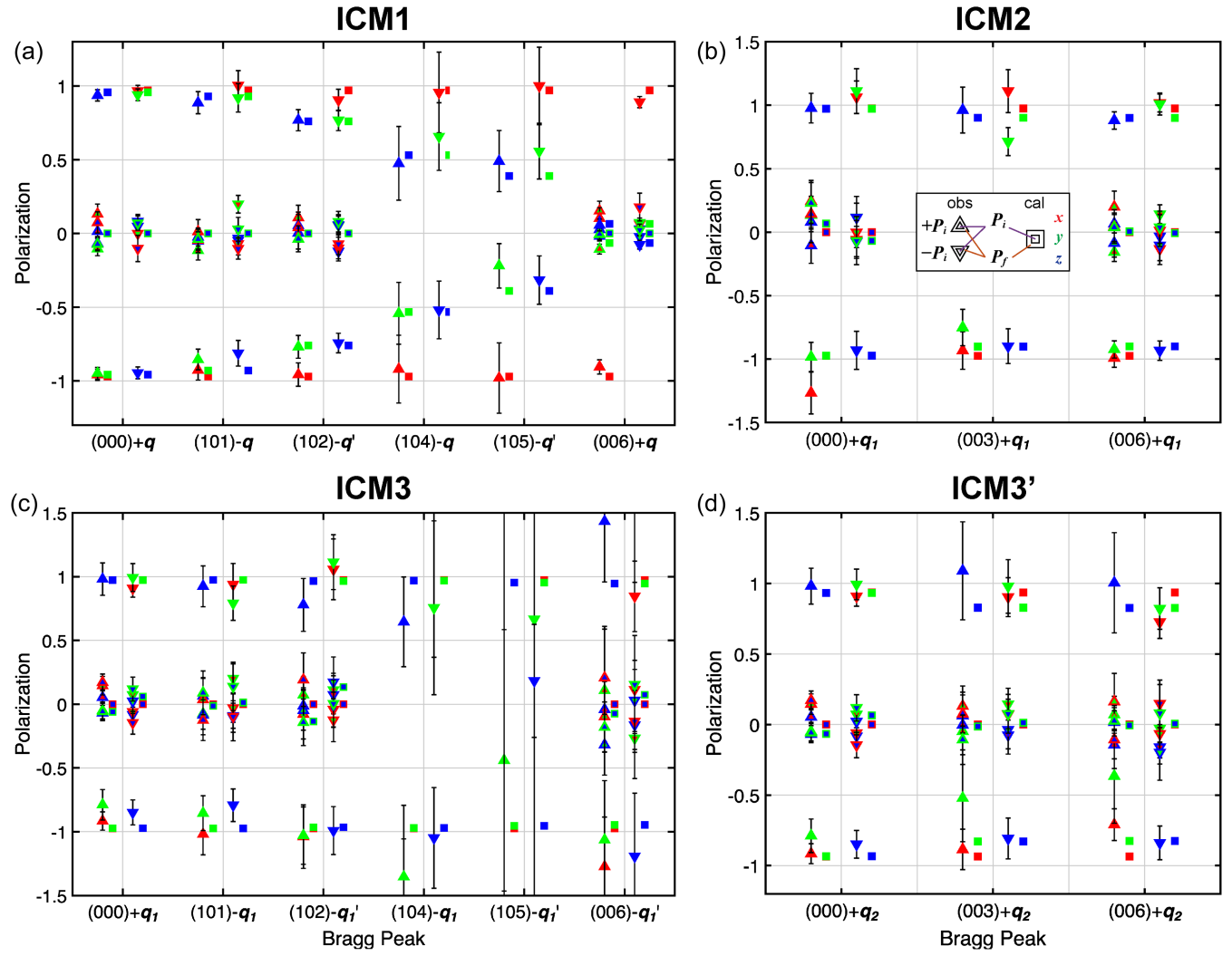


FIG. 5. (a)–(d) The polarization matrix elements for different magnetic satellite peaks and the corresponding model intensities from the models depicted in Figs. 1(f)–1(h) for (a) ICM1, (b) ICM2, (c) the $H0L$ plane of ICM3, and (d) the HHL plane of ICM3. The incident and outgoing neutron polarizations are indicated with the inner and outer marker color, respectively. Red, green, and blue correspond to x , y , and z , respectively. The upward and downward triangles indicate a positive and negative incident neutron polarization, respectively. Measured matrix elements are indicated with triangles, while the elements calculated from the model are indicated with squares. All error bars include both statistical and estimated systematic uncertainties.

TABLE I. The calculated and observed polarization matrix elements for different magnetic satellite peaks in ICM1. The first column lists the measured diffraction peak. The second column lists the incident neutron polarization direction. The next three columns list the modeled polarization matrix elements, and the final three list the measured polarization matrix elements with no polarization efficiency correction applied. All error bars include both statistical and estimated systematic uncertainties.

(H,K,L)	P_i	P_{fx}^{calc}	P_{fy}^{calc}	P_{fz}^{calc}	P_{fx}^{obs}	P_{fy}^{obs}	P_{fz}^{obs}
(0.108,0,0.053)	+x	-0.97	0	0	-0.96(4)	-0.10(5)	0.01(5)
	+y	0	-0.957	0	0.13(7)	-0.95(4)	-0.07(4)
	+z	0	0	0.957	0.07(7)	-0.07(5)	0.94(4)
	-x	0.97	0	0	0.97(4)	0.08(4)	0.05(4)
	-y	0	0.957	0	-0.00(9)	0.94(4)	0.08(5)
	-z	0	0	-0.957	-0.10(9)	0.07(5)	-0.95(4)
(0.892,0,0.947)	+x	-0.97	0	0	-0.93(7)	-0.11(7)	-0.02(6)
	+y	0	-0.93	0	0.01(8)	-0.85(7)	-0.05(6)
	+z	0	0	0.93	-0.05(8)	-0.04(7)	0.89(8)
	-x	0.97	0	0	1.00(10)	0.20(6)	-0.05(6)
	-y	0	0.93	0	-0.07(7)	0.92(10)	-0.03(8)
	-z	0	0	-0.93	-0.10(7)	0.03(8)	-0.81(9)
(0.892,0,2.053)	+x	-0.97	0	0	-0.96(8)	-0.04(7)	0.00(6)
	+y	0	-0.76	0	0.11(8)	-0.77(8)	0.05(9)
	+z	0	0	0.76	0.04(9)	-0.04(9)	0.77(7)
	-x	0.97	0	0	0.91(7)	0.05(6)	-0.12(7)
	-y	0	0.76	0	-0.10(7)	0.77(7)	0.06(7)
	-z	0	0	-0.76	-0.07(7)	0.08(7)	-0.74(7)
(0.892,0,3.947)	+x	-0.97			-0.9(2)		
	+y		-0.532			-0.5(2)	
	+z			0.532			0.5(2)
	-x	0.97			1.0(3)		
	-y		0.532			0.7(2)	
	-z			-0.532			-0.5(2)
(0.892,0,5.053)	+x	-0.97			-1.0(2)		
	+y		-0.39			-0.2(2)	
	+z			0.39			0.5(2)
	-x	0.97			1.0(3)		
	-y		0.39			0.6(2)	
	-z			-0.39			-0.3(2)
(0.108,0,6.053)	+x	-0.97	0	0	-0.90(5)	-0.11(3)	-0.02(3)
	+y	0	-0.065	0	0.15(6)	-0.01(4)	0.03(4)
	+z	0	0	0.065	0.10(7)	-0.00(4)	0.06(4)
	-x	0.97	0	0	0.89(4)	0.03(3)	0.03(3)
	-y	0	0.065	0	0.1(1)	0.07(3)	-0.02(4)
	-z	0	0	-0.065	0.18(9)	0.02(4)	-0.08(3)

TABLE II. The calculated and observed polarization matrix elements for different magnetic satellite peaks in ICM2. The first column lists the measured diffraction peak. The second column lists the incident neutron polarization direction. The next three columns list the modeled polarization matrix elements, and the final three list the measured polarization matrix elements with no polarization efficiency correction applied. All error bars include both statistical and estimated systematic uncertainties.

(H,K,L)	P_i	P_{fx}^{calc}	P_{fy}^{calc}	P_{fz}^{calc}	P_{fx}^{obs}	P_{fy}^{obs}	P_{fz}^{obs}
(0.076,0.032,0.01)	+x	-0.975	0	0	-1.3(2)	0.1(1)	-0.1(1)
	+y	0	-0.973	0.068	0.2(2)	-1.0(1)	0.1(2)
	+z	0	0.068	0.973	0.1(1)	0.2(2)	1.0(1)
	-x	0.975	0	0	1.1(1)	0.1(1)	-0.1(1)
	-y	0	0.973	-0.068	-0.0(1)	1.1(2)	0.1(2)
	-z	0	-0.068	-0.973	-0.1(1)	-0.1(2)	-0.9(2)
(0.076,0.032,3.01)	+x	-0.975			-0.9(1)		
	+y		-0.902			-0.8(1)	
	+z			0.902			1.0(2)
	-x	0.975			1.1(2)		
	-y		0.902			0.7(1)	
	-z			-0.902			-0.9(1)
(0.076,0.032,6.01)	+x	-0.975	0	0	-0.99(7)	-0.16(7)	0.07(7)
	+y	0	-0.9	0.006	0.2(1)	-0.92(7)	-0.1(1)
	+z	0	0.006	0.9	0.0(1)	0.0(1)	0.88(7)
	-x	0.975	0	0	1.02(7)	0.14(7)	-0.03(7)
	-y	0	0.9	-0.006	-0.1(1)	1.01(9)	-0.1(1)
	-z	0	-0.006	-0.9	0.0(1)	0.0(1)	-0.93(8)

TABLE III. The calculated and observed polarization matrix elements for different magnetic satellite peaks in ICM3 in the $H0L$ plane. The first column lists the measured diffraction peak. The second column lists the incident neutron polarization direction. The next three columns list the modeled polarization matrix elements, and the final three list the measured polarization matrix elements with no polarization efficiency correction applied. All error bars include both statistical and estimated systematic uncertainties.

(H,K,L)	P_i	P_{fx}^{calc}	P_{fy}^{calc}	P_{fz}^{calc}	P_{fx}^{obs}	P_{fy}^{obs}	P_{fz}^{obs}
(0.11,0,0.037)	+x	-0.975	0	0	-0.92(7)	-0.05(7)	0.05(6)
	+y	0	-0.973	-0.06	0.14(7)	-0.8(1)	-0.07(6)
	+z	0	-0.06	0.973	0.17(7)	-0.06(6)	1.0(1)
	-x	0.975	0	0	0.91(7)	0.07(6)	-0.08(6)
	-y	0	0.973	0.06	-0.06(9)	1.0(1)	0.02(9)
	-z	0	0.06	-0.973	-0.15(9)	0.12(9)	-0.85(10)
(0.89,0,0.963)	+x	-0.975	0	0	-1.0(2)	-0.1(1)	0.1(1)
	+y	0	-0.975	-0.014	0.0(2)	-0.9(1)	-0.1(2)
	+z	0	-0.014	0.975	-0.1(2)	0.1(2)	0.9(2)
	-x	0.975	0	0	0.9(2)	0.2(1)	-0.09(10)
	-y	0	0.975	0.014	-0.0(1)	0.8(1)	-0.1(2)
	-z	0	0.014	-0.975	-0.1(1)	0.1(2)	-0.8(1)
(0.89,0,2.037)	+x	-0.975	0	0	-1.0(2)	0.1(1)	-0.0(1)
	+y	0	-0.965	-0.135	-0.1(2)	-1.0(2)	-0.0(2)
	+z	0	-0.135	0.965	0.2(2)	-0.1(2)	0.8(2)
	-x	0.975	0	0	1.1(2)	0.1(1)	0.1(1)
	-y	0	0.965	0.135	-0.1(2)	1.1(2)	0.2(2)
	-z	0	0.135	-0.965	-0.0(1)	0.0(2)	-1.0(2)
(0.89,0,3.963)	+x	-0.975			-2(1)		
	+y		-0.97			-1.4(6)	
	+z			0.97			0.6(4)
	-x	0.975			2(2)		
	-y		0.97			0.8(7)	
	-z			-0.97			-1.0(4)

(Table continued)

TABLE III. (*Continued*)

(H,K,L)	P_i	p_{fx}^{calc}	p_{fy}^{calc}	p_{fz}^{calc}	p_{fx}^{obs}	p_{fy}^{obs}	p_{fz}^{obs}
(0.89,0,5.037)	$+x$	−0.975	−0.954	0.954	0(10)	0(1)	10(50)
	$+y$						
	$+z$						
	$-x$	0.975	0.954	−0.954	−10(30)	1(2)	0.2(4)
	$-y$						
	$-z$						
(−0.11,0,6.037)	$+x$	−0.975	0	0	−1.3(4)	0.1(5)	−0.0(3)
	$+y$						
	$+z$						
	$-x$	0.975	0	0	0.8(3)	−0.3(3)	−0.2(2)
	$-y$						
	$-z$						
		0	−0.947	−0.074	−0.1(3)	−1.1(5)	−0.3(2)
		0	−0.074	0.947	0.2(4)	−0.2(2)	1.4(5)
		0.975	0	0	0.8(3)	−0.3(3)	−0.2(2)
		0	0.947	0.074	0.1(2)	1.8(8)	0.0(2)
		0	0.074	−0.947	−0.1(2)	0.2(4)	−1.2(5)

TABLE IV. The calculated and observed polarization matrix elements for different magnetic satellite peaks in ICM3 in the *HKL* plane. The first column lists the measured diffraction peak. The second column lists the incident neutron polarization direction. The next three columns list the modeled polarization matrix elements, and the final three list the measured polarization matrix elements with no polarization efficiency correction applied. All error bars include both statistical and estimated systematic uncertainties.

(H,K,L)	P_i	p_{fx}^{calc}	p_{fy}^{calc}	p_{fz}^{calc}	p_{fx}^{obs}	p_{fy}^{obs}	p_{fz}^{obs}
(0.06,0.06,0)	$+x$	−0.935	0	0	−0.92(7)	−0.05(7)	0.05(6)
	$+y$						
	$+z$						
	$-x$	0.935	0	0	0.91(7)	0.07(6)	−0.08(6)
	$-y$						
	$-z$						
(0.06,0.06,3)	$+x$	−0.935	0	0	−0.9(1)	−0.0(1)	−0.0(1)
	$+y$						
	$+z$						
	$-x$	0.935	0	0	0.9(1)	0.1(1)	−0.1(1)
	$-y$						
	$-z$						
(0.06,0.06,6)	$+x$	−0.935	0	0	−0.7(1)	0.1(1)	−0.14(10)
	$+y$						
	$+z$						
	$-x$	0.935	0	0	0.7(1)	−0.0(1)	−0.2(1)
	$-y$						
	$-z$						
		0	−0.826	−0.006	0.2(2)	−0.4(3)	0.0(1)
		0	−0.006	0.826	−0.1(2)	0.0(1)	1.0(4)
		0.935	0	0	0.7(1)	−0.0(1)	−0.2(1)
		0	0.826	0.006	−0.1(2)	0.8(1)	−0.2(2)
		0	0.006	−0.826	0.1(2)	0.1(2)	−0.8(1)

- [1] George Grüner, *The dynamics of charge-density waves*, *Rev. Mod. Phys.* **60**, 1129 (1988).
- [2] George Gruner, *Density Waves in Solids* (Addison-Wesley Pub. Co., Advanced Book Program, Reading, MA, 1994).
- [3] Kotaro Shimizu, Shun Okumura, Yasuyuki Kato, and Yukitoshi Motome, *Spin Moiré engineering of topological magnetism and emergent electromagnetic fields*, *Phys. Rev. B* **103**, 184421 (2021).

- [4] Ryo Ozawa, Satoru Hayami, Kipton Barros, Gia-Wei Chern, Yukitoshi Motome, and Cristian D. Batista, *Vortex crystals with chiral stripes in itinerant magnets*, *J. Phys. Soc. Jpn.* **85**, 103703 (2016).
- [5] Ryo Ozawa, Satoru Hayami, and Yukitoshi Motome, *Zero-field skyrmions with a high topological number in itinerant magnets*, *Phys. Rev. Lett.* **118**, 147205 (2017).
- [6] Zhentao Wang, Ying Su, Shi-Zeng Lin, and Cristian D. Batista, *Skyrmion crystal from RKKY interaction mediated by 2D electron gas*, *Phys. Rev. Lett.* **124**, 207201 (2020).
- [7] Satoru Hayami and Yukitoshi Motome, *Noncoplanar multiple-*Q* spin textures by itinerant frustration: Effects*

- of single-ion anisotropy and bond-dependent anisotropy, *Phys. Rev. B* **103**, 054422 (2021).
- [8] Takashi Kurumaji, Nisarga Paul, Shiang Fang, Paul M. Neves, Mingu Kang, Jonathan S. White, Taro Nakajima, David Graf, Linda Ye, Mun K. Chan *et al.*, *Electronic commensuration of a spin Moiré superlattice in a layered magnetic semimetal*, *Sci. Adv.* **11**, eadu6686 (2025).
- [9] N. D. Andriushin, J. Muller, N. S. Pavlovskii, J. Grumbach, S. Granovsky, Y. V. Tymoshenko, O. Zaharko, A. Ivanov, J. Ollivier, M. Doerr *et al.*, *Reentrant multiple- q magnetic order and a “spin meta-cholesteric” phase in $\text{Sr}_3\text{Fe}_2\text{O}_7$* , *npj Quantum Mater.* **9**, 84 (2024).
- [10] J. Green, Harry W. T. Morgan, M. Mandigo-Stoba, W. T. Laderer, K.-Y. Wey, A. G. Prado, C. Jozwiak, A. Bostwick, E. Rotenberg, C. Gutiérrez, A. N. Alexandrova, and N. Ni, *Mapping the three-dimensional fermiology of the triangular lattice magnet EuAg_4Sb_2* , *Phys. Rev. B* **111**, 085139 (2025).
- [11] S. Malick, H. Świątek, J. Bławat, J. Singleton, and T. Klimczuk, *Large magnetoresistance and first-order phase transition in antiferromagnetic single-crystalline EuAg_4Sb_2* , *Phys. Rev. B* **110**, 165149 (2024).
- [12] J. Green, Arpit Arora, Madalynn Marshall, Wanfei Shan, Péter Udvarhelyi, Zachary Morgan, Prineha Narang, Huibo Cao, and Ni Ni, *Robust spin-moiré-superlattice-driven gap opening in EuAg_4Sb_2 under in-plane magnetic field*, *Nano Lett.* **25**, 13723 (2025).
- [13] Dmitry Solenov, Dmitry Mozysky, and Ivar Martin, *Chirality waves in two-dimensional magnets*, *Phys. Rev. Lett.* **108**, 096403 (2012).
- [14] Keita Hamamoto, Motohiko Ezawa, and Naoto Nagaosa, *Quantized topological Hall effect in skyrmion crystal*, *Phys. Rev. B* **92**, 115417 (2015).
- [15] Gustau Catalan, J. Seidel, Ramamoorthy Ramesh, and James F. Scott, *Domain wall nanoelectronics*, *Rev. Mod. Phys.* **84**, 119 (2012).
- [16] Pankaj Sharma, Qi Zhang, Daniel Sando, Chi Hou Lei, Yunya Liu, Jiangyu Li, Valanoor Nagarajan, and Jan Seidel, *Nonvolatile ferroelectric domain wall memory*, *Sci. Adv.* **3**, e1700512 (2017).
- [17] Angela Haykal, Johanna Fischer, Waqar Akhtar, J.-Y. Chaudhary, D. Sando, A. Finco, Florian Godel, Y. A. Birkhölzer, C. Carrétéro, Nicolas Jaouen *et al.*, *Antiferromagnetic textures in BiFeO_3 controlled by strain and electric field*, *Nat. Commun.* **11**, 1704 (2020).
- [18] A. S. Sukhanov, P. Vir, A. Heinemann, S. E. Nikitin, D. Kriegner, H. Borrmann, C. Shekhar, C. Felser, and D. S. Inosov, *Giant enhancement of the skyrmion stability in a chemically strained helimagnet*, *Phys. Rev. B* **100**, 180403 (R) (2019).
- [19] W. L. N. C. Liyanage, Nan Tang, Rebecca L. Dally, Elizabeth J. Quigley, C. Charlotte Buchanan, Guo-Jiun Shu, Nicholas P. Butch, Kathryn Krycka, Markus Bleuel, Julie A. Borchers *et al.*, *Skyrmion lattice formation and destruction mechanisms probed with TR-SANS*, *Nanoscale* **16**, 10715 (2024).
- [20] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/bjq3-py7l> for detailed information on the domain and phase characterization, as well as the polarized neutron scattering, which includes Refs. [21,22].
- [21] S. Parsons, *Introduction to twinning*, *Biol. Crystallogr.* **59**, 1995 (2003).
- [22] Kotaro Shimizu, Shun Okumura, Yasuyuki Kato, and Yukitoshi Motome, *Phase degree of freedom and topology in multiple- Q spin textures*, *Phys. Rev. B* **105**, 224405 (2022).
- [23] M. Ramazanoglu, W. Ratcliff, H. T. Yi, A. A. Sirenko, S.-W. Cheong, and V. Kiryukhin, *Giant effect of uniaxial pressure on magnetic domain populations in multiferroic bismuth ferrite*, *Phys. Rev. Lett.* **107**, 067203 (2011).
- [24] C. Dhital, Z. Yamani, W. Tian, J. Zeretsky, A. S. Sefat, Z. Wang, R. J. Birgeneau, and S. D. Wilson, *Effect of uniaxial strain on the structural and magnetic phase transitions in BaFe_2As_2* , *Phys. Rev. Lett.* **108**, 087001 (2012).
- [25] Joonbum Park, Hideaki Sakai, Andrew P. Mackenzie, and Clifford W. Hicks, *Effect of uniaxial stress on the magnetic phases of CeAuSb_2* , *Phys. Rev. B* **98**, 024426 (2018).
- [26] J.-W. Kim, S. H. Chun, Y. Choi, B. J. Kim, M. H. Upton, and P. J. Ryan, *Controlling symmetry of spin-orbit entangled pseudospin state through uniaxial strain*, *Phys. Rev. B* **102**, 054420 (2020).
- [27] Masaki Gen, Rina Takagi, Yoshito Watanabe, Shunsuke Kitou, Hajime Sagayama, Naofumi Matsuyama, Yoshimitsu Kohama, Akihiko Ikeda, Yoshichika Ōnuki, Takashi Kurumaji *et al.*, *Rhombic skyrmion lattice coupled with orthorhombic structural distortion in EuAl_4* , *Phys. Rev. B* **107**, L020410 (2023).
- [28] A. Chacon, A. Bauer, T. Adams, F. Rucker, G. Brandl, R. Georgii, M. Garst, and C. Pfleiderer, *Uniaxial pressure dependence of magnetic order in MnSi* , *Phys. Rev. Lett.* **115**, 267202 (2015).
- [29] Y. Nii, T. Nakajima, A. Kikkawa, Y. Yamasaki, K. Ohishi, J. Suzuki, Y. Taguchi, T. Arima, Y. Tokura, and Y. Iwasa, *Uniaxial stress control of skyrmion phase*, *Nat. Commun.* **6**, 8539 (2015).
- [30] Taro Nakajima, Victor Ukleev, Kazuki Ohishi, Hiroshi Oike, Fumitaka Kagawa, Shin-ichiro Seki, Kazuhisa Kakurai, Yoshinori Tokura, and Taka-hisa Arima, *Uniaxial-stress effects on helimagnetic orders and skyrmion lattice in Cu_2OSeO_3* , *J. Phys. Soc. Jpn.* **87**, 094709 (2018).
- [31] Birgit Gerke, Christian Schwickert, Stanislav S. Stoyko, Mansura Khatun, Arthur Mar, and Rainer Pöttgen, *Magnetic hyperfine field splitting in EuAg_4As_2 and EuAg_4Sb_2* , *Solid State Sci.* **20**, 65 (2013).
- [32] P. Rhodes and E. P. Wohlfarth, *The effective Curie-Weiss constant of ferromagnetic metals and alloys*, *Proc. R. Soc. A* **273**, 247 (1963).
- [33] E. M. Forgan, *Complex structures and simple ideas in modulated antiferromagnets*, *J. Magn. Magn. Mater.* **104–107**, 1485 (1992).
- [34] Satoru Hayami and Ryota Yambe, *Stabilization mechanisms of magnetic skyrmion crystal and multiple- q states based on momentum-resolved spin interactions*, *Mater. Today Quantum* **3**, 100010 (2024).
- [35] E. M. Forgan, E. P. Gibbons, K. A. McEwen, and D. Fort, *Observation of a quadruple- q magnetic structure in neodymium*, *Phys. Rev. Lett.* **62**, 470 (1989).
- [36] Satoru Hayami, Ryo Ozawa, and Yukitoshi Motome, *Effective bilinear-biquadratic model for noncoplanar ordering in itinerant magnets*, *Phys. Rev. B* **95**, 224424 (2017).

- [37] Souvik Paul, Soumyajyoti Halder, Stephan von Malottki, and Stefan Heinze, *Role of higher-order exchange interactions for skyrmion stability*, *Nat. Commun.* **11**, 4756 (2020).
- [38] M. Colarieti-Tosti, S. I. Simak, R. Ahuja, L. Nordström, O. Eriksson, D. Åberg, S. Edvardsson, and M. S. S. Brooks, *Origin of magnetic anisotropy of Gd metal*, *Phys. Rev. Lett.* **91**, 157201 (2003).
- [39] M. Blanco-Rey, R. Castrillo-Bodero, K. Ali, P. Gargiani, F. Bertran, P. M. Sheverdyaeva, J. E. Ortega, L. Fernández, and F. Schiller, *Effect of the valence state on the band magnetocrystalline anisotropy in two-dimensional rare-earth/noble-metal compounds*, *Phys. Rev. Res.* **4**, 013237 (2022).
- [40] Z.-C. Wang, E. Been, J. Gaudet, Gadeer Matook A. Alqasseri, K. Fruhling, X. Yao, U. Stuhr, Q. Zhu, Z. Ren, Y. Cui, C. Jia, B. Moritz, S. Chowdhury, T. Devereaux, and F. Tafti, *Anisotropy of the magnetic and transport properties of EuZn₂As₂*, *Phys. Rev. B* **105**, 165122 (2022).
- [41] Ai Nakamura, Taro Uejo, Fuminori Honda, Tetsuya Takeuchi, Hisatomo Harima, Etsuji Yamamoto, Yoshinori Haga, Kazuyuki Matsubayashi, Yoshiya Uwatoko, Masato Hedo *et al.*, *Transport and magnetic properties of EuAl₄ and EuGa₄*, *J. Phys. Soc. Jpn.* **84**, 124711 (2015).
- [42] R. Takagi, N. Matsuyama, V. Ukleev, L. Yu, J. S. White, S. Francoual, J. R. L. Mardegan, S. Hayami, H. Saito, K. Kaneko, K. Ohishi, Y. Ōnuki, T. Arima, Y. Tokura, T. Nakajima, and S. Seki, *Square and rhombic lattices of magnetic skyrmions in a centrosymmetric binary compound*, *Nat. Commun.* **13**, 1472 (2022).
- [43] Paul M. Neves, Takashi Kurumaji, Joshua P. Wakefield, Chi Ian Jess Ip, Robert Cubitt, Satoru Hayami, Jonathan S. White, and Joseph G. Checkelsky, *Cascade of spin moiré superlattices with in-plane field in triangular lattice semi-metal EuAg₄Sb₂*, *ACS Nano* **20**, 14029 (2026).
- [44] C. D. Dewhurst, *Graphical reduction and analysis small-angle neutron scattering program: GRASP*, *J. Appl. Crystallogr.* **56**, 1595 (2023).
- [45] Paul M. Neves and Jonathan S. White, *GRASP integrated 3D plotter: GRIP*, *J. Appl. Crystallogr.* **57**, 2030 (2024).
- [46] Dirk Honecker and Andreas Michels, *Theory of magnetic small-angle neutron scattering of two-phase ferromagnets*, *Phys. Rev. B* **87**, 224426 (2013).
- [47] Sebastian Mühlbauer, Dirk Honecker, Élio A. Périgo, Frank Bergner, Sabrina Disch, André Heinemann, Sergey Erokhin, Dmitry Berkov, Chris Leighton, Morten Ring Eskildsen *et al.*, *Magnetic small-angle neutron scattering*, *Rev. Mod. Phys.* **91**, 015004 (2019).
- [48] K. H. Andersen, R. Chung, V. Guillard, H. Humblot, D. Jullien, E. Lelievre-Berna, A. Petoukhov, and F. Tasset, *First results from Tyrex, the new polarized-3He filling station at ILL*, *Physica (Amsterdam)* **356B**, 103 (2005).
- [49] M. Batz, S. Baeßler, W. Heil, E. W. Otten, D. Rudersdorf, J. Schmiedeskamp, Y. Sobolev, and M. Wolf, *3He spin filter for neutrons*, *J. Res. Natl. Inst. Stand. Technol.* **110**, 293 (2005).
- [50] A. R. Wildes, *Scientific reviews: Neutron polarization analysis corrections made easy*, *Neutron News* **17**, 17 (2006).
- [51] Andrew T Boothroyd, *Principles of Neutron Scattering from Condensed Matter* (Oxford University Press, New York, 2020).
- [52] F. Tasset, P. J. Brown, E. Lelievre-Berna, T. Roberts, S. Pujol, J. Allibon, and E. Bourgeat-Lami, *Spherical neutron polarimetry with Cryopad-II*, *Physica (Amsterdam)* **267B–268B**, 69 (1999).
- [53] Navid Qureshi, *Mag2Pol: A program for the analysis of spherical neutron polarimetry, flipping ratio and integrated intensity data*, *Appl. Crystallogr.* **52**, 175 (2019).
- [54] N. Giles-Donovan, N. Qureshi, R. D. Johnson, L. Y. Zhang, S.-W. Cheong, S. Cochran, and C. Stock, *Imitation of spin density wave order in Cu₃Nb₂O₈*, *Phys. Rev. B* **102**, 024414 (2020).
- [55] 10.5291/ILL-DATA.DIR-324, 10.5291/ILL-DATA.5-54-408, and 10.5291/ILL-DATA.5-71-2.
- [56] Paul Neves, *Replication data for General approach to solving spin moiré superstructures*, *Harvard Dataverse*, 2026, 10.7910/DVN/VZQWIL.